BEING PRO-ISOMORPHIC TO A DELIGNE SYSTEM IS LOCAL ON AN OPEN COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G4T (sources/stacks-duality/pinned-original-source.tex line 8358).

Allowed standard prerequisites (editorial).

- Deligne systems and their restrictions to opens. Same chapter, the definition of a Deligne system preceding this lemma, invoked in the statement and at proof lines 8384-8386. Fixes the property being tested locally.
- The colimit of Hom out of a Deligne system computes Hom on the open. Same chapter, Lemma labelled lemma-lift-map, cited at proof lines 8408-8412. Makes all but the middle horizontal arrow of the diagram an isomorphism.
- Mayer-Vietoris for Hom and essential constancy along distinguished triangles. Cohomology, Lemma labelled cohomology-lemma-mayer-vietoris-hom, cited at proof lines 8404-8406, and Cohomology, Lemma labelled cohomology-lemma-exact-sequence-j-star with Derived Categories, Lemma labelled derived-lemma-essentially-constant-2-out-of-3, cited at proof lines 8388-8392. Give the exactness of the vertical sequences and the essential constancy of the system on U.
- Existence of a Deligne system with prescribed constant value, and pro-isomorphism through colimits of Hom. Same chapter, Lemma labelled lemma-extension-by-zero, cited at proof lines 8418-8420, and Categories, Remark labelled categories-remark-pro-category-copresheaves, cited at proof lines 8422-8425. Concludes that the given system is pro-isomorphic to a Deligne system.

Source reference. Original lemma 08BV: sources/stacks-chow/tags.txt, source lines 10580–10580. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8388 of the counted proof. It is

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Source reference. Original lemma 0G3A: sources/stacks-chow/tags.txt, source lines 20085–20085. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8389 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08BW: sources/stacks-chow/tags.txt, source lines 10581–10581. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8412 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G2H: sources/stacks-duality/pinned-original-source.tex, source lines 7920–7920. Same-chapter pointer into the author environment lemma at source line 7920; invoked at proof line 8415 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4L: sources/stacks-duality/pinned-original-source.tex, source lines 8103–8103. Same-chapter pointer into the author environment lemma at source line 8103; invoked at proof line 8420 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0G2X: sources/stacks-chow/tags.txt, source lines 20072–20072. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8424 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $j : U \to X$ be an open immersion of Noetherian schemes. Let (K_n) be an inverse system in $D^b_{Coh}(\mathcal{O}_X)$. Let $X = W_1 \cup \ldots \cup W_r$ be an open covering. The following are equivalent*

(1) *(K_n) is pro-isomorphic to a Deligne system,*
(2) *for each i the restriction $(K_n|_{W_i})$ is pro-isomorphic to a Deligne system with respect to the open immersion $U \cap W_i \to W_i$.*

Proof. By induction on r. If $r = 1$ then the result is clear. Assume $r > 1$. Set $V = W_1 \cup \ldots \cup W_{r-1}$. By induction we see that $(K_n|_V)$ is a Deligne system. This reduces us to the discussion in the next paragraph.

Assume $X = V \cup W$ is an open covering and $(K_n|_W)$ and $(K_n|_V)$ are pro-isomorphic to Deligne systems. We have to show that (K_n) is pro-isomorphic to a Deligne system. Observe that $(K_n|_{V\cap W})$ is pro-isomorphic to a Deligne system (it follows immediately from the construction of Deligne systems that restrictions to opens preserves them). In particular the pro-systems $(K_n|_{U\cap V})$, $(K_n|_{U\cap W})$, and $(K_n|_{U\cap V\cap W})$ are essentially constant. It follows from the distinguished triangles in Cohomology, Lemma 08BV and Derived Categories, Lemma 0G3A that $(K_n|_U)$ is essentially constant. Denote $K \in D^b_{Coh}(\mathcal{O}_U)$ the value of this system. Let L be an

object of $D^b_{Coh}(\mathcal{O}_X)$. Consider the diagram

$$\begin{array}{ccc}
\operatorname{colim}\operatorname{Ext}^{-1}(K_n|_V, L|_V) \oplus \operatorname{colim}\operatorname{Ext}^{-1}(K_n|_W, L|_W) & \longrightarrow & \operatorname{Ext}^{-1}(K|_{U\cap V}, L|_{U\cap V}) \oplus \operatorname{Ext}^{-1}(K|_{U\cap W}, L|_{U\cap W}) \\
\downarrow & & \downarrow \\
\operatorname{colim}\operatorname{Ext}^{-1}(K_n|_{V\cap W}, L|_{V\cap W}) & \longrightarrow & \operatorname{Ext}^{-1}(K|_{U\cap V\cap W}, L|_{U\cap V\cap W}) \\
\downarrow & & \downarrow \\
\operatorname{colim}\operatorname{Hom}(K_n, L) & \longrightarrow & \operatorname{Hom}(K|_U, L|_U) \\
\downarrow & & \downarrow \\
\operatorname{colim}\operatorname{Hom}(K_n|_V, L|_V) \oplus \operatorname{colim}\operatorname{Hom}(K_n|_W, L|_W) & \longrightarrow & \operatorname{Hom}(K|_{U\cap V}, L|_{U\cap V}) \oplus \operatorname{Hom}(K|_{U\cap W}, L|_{U\cap W}) \\
\downarrow & & \downarrow \\
\operatorname{colim}\operatorname{Hom}(K_n|_{V\cap W}, L|_{V\cap W}) & \longrightarrow & \operatorname{Hom}(K|_{U\cap V\cap W}, L|_{U\cap V\cap W})
\end{array}$$

The vertical sequences are exact by Cohomology, Lemma 08BW and the fact that filtered colimits are exact. All horizontal arrows except for the middle one are isomorphisms by Lemma 0G2H and the fact that pro-isomorphic systems have the same colimits. Hence the middle one is an isomorphism too by the 5-lemma. It follows that (K_n) is pro-isomorphic to a Deligne system for K. Namey, if (K'_n) is a Deligne system whose restriction to U has constant value K (which exists by Lemma 0G4L), then we have $\operatorname{colim}\operatorname{Hom}_X(K'_n, L) = \operatorname{Hom}_U(K, L|_U)$ as well. Hence the pro-systems (K_n) and (K'_n) are pro-isomorphic by Categories, Remark 0G2X. □